Text Embeddings and Semantic Similarity

1. What are Embeddings?

An embedding is a numerical representation of an item such as text. An embedding model maps text into a vector in a multidimensional space. Texts with related meanings can produce vectors that are close according to a chosen similarity measure.

2. From Text to Vectors

A document chunk can be passed to an embedding model to produce a vector. The same model is used to transform a user query into a vector. The query vector can then be compared with stored document vectors to identify semantically related content.

3. Semantic Search

Semantic search retrieves information based on meaning rather than requiring an exact keyword match. For example, a question about finding similar documents can retrieve a passage describing vector similarity even if the passage does not use the exact wording of the question.

4. Similarity

Vector search commonly ranks stored vectors according to a similarity or distance function. Cosine similarity is one commonly used measure for comparing the direction of two vectors. The resulting ranking can be used to select the top candidate document chunks.

5. What Embeddings Do Not Do

Embeddings do not contain a guaranteed textual answer to every question. They provide representations used for comparison and retrieval. If the knowledge base does not contain the required information, semantic search cannot reliably create that missing fact.

6. Use in RAG

In a RAG application, embeddings are typically created for document chunks during indexing and for user queries at search time. The nearest relevant chunks are then passed to the language model as context for answer generation.